# covid | the 10 largest locations without Florida | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

Horizon -1
Horizon 0
Horizon 1
Horizon 2
Horizon 3
Feedback delay 0
Feedback delay 1
Feedback re-run
Jun 2025 to Sep 2026
Up to peak (weeks -7 to 0)
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 10 locations
Horizon 0 | delay 0 | 10 locations
Horizon 1 | delay 0 | 10 locations
Horizon 2 | delay 0 | 10 locations
Horizon 3 | delay 0 | 10 locations
Horizon -1 | delay 1 | 10 locations
Horizon 0 | delay 1 | 10 locations
Horizon 1 | delay 1 | 10 locations
Horizon 2 | delay 1 | 10 locations
Horizon 3 | delay 1 | 10 locations
Horizon -1 | re-run | 10 locations
Horizon 0 | re-run | 10 locations
Horizon 1 | re-run | 10 locations
Horizon 2 | re-run | 10 locations
Horizon 3 | re-run | 10 locations
Calibration error
Mean quantile loss (counts per 100k residents)
laptive rate)
ıdaptive rate, leak 0.1)
Leak chosen online
Leaky MultiQT, leak chosen on the last 20 rows (leaks 0, 0.02, 0.05, 0.1, 0.2, 0.3, 0.5)
Leaky MultiQT, leak bank: learned common weights + sharing (H=8)
Rate-grid experts
Rate grid: equal
Rate grid: learned, no sharing (H=8)
Rate grid: learned, no sharing, common weights (H=8)
Weekly starts: immediate admission
Weekly starts: equal (m=0.03)
Weekly starts: most recent 8, learned + sharing (m=0.03, H=8)
Weekly starts: most recent 8, slow + fast, learned + sharing (m=0.03 + 0.3, H=8)
Score-based experts
Score experts: learned weights per level (H=8)
Power scale
Studentized full base ((1 + median)^0.5, adaptive rate)
Power scale: rate grid, learned, common weights (H=8)
Power scale: most recent 8 starts, learned + sharing (m=0.03, H=8)
Spread scale
Studentized full base (spread per level, lambda=0.4, adaptive rate)
Spread scale: rate grid, learned, common weights (H=8)
Spread scale: most recent 8 starts, learned + sharing (m=0.03, H=8)
Sign-flip experts
Sign-flip experts: base, power-scale tracker, mirror (H=8)
Sign-flip experts: base, level-RMS tracker, mirror (H=8)
Sign-flip experts: base, base-width tracker, mirror (H=8)
Two-phase tracker
Two-phase power sc
Two-phase power sc

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Every tracker learns at the adaptive rate (no tuning); the same settings for every feedback design.

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Horizon -1 | delay 1 | 10 locations
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Horizon 3 | delay 1 | 10 locations
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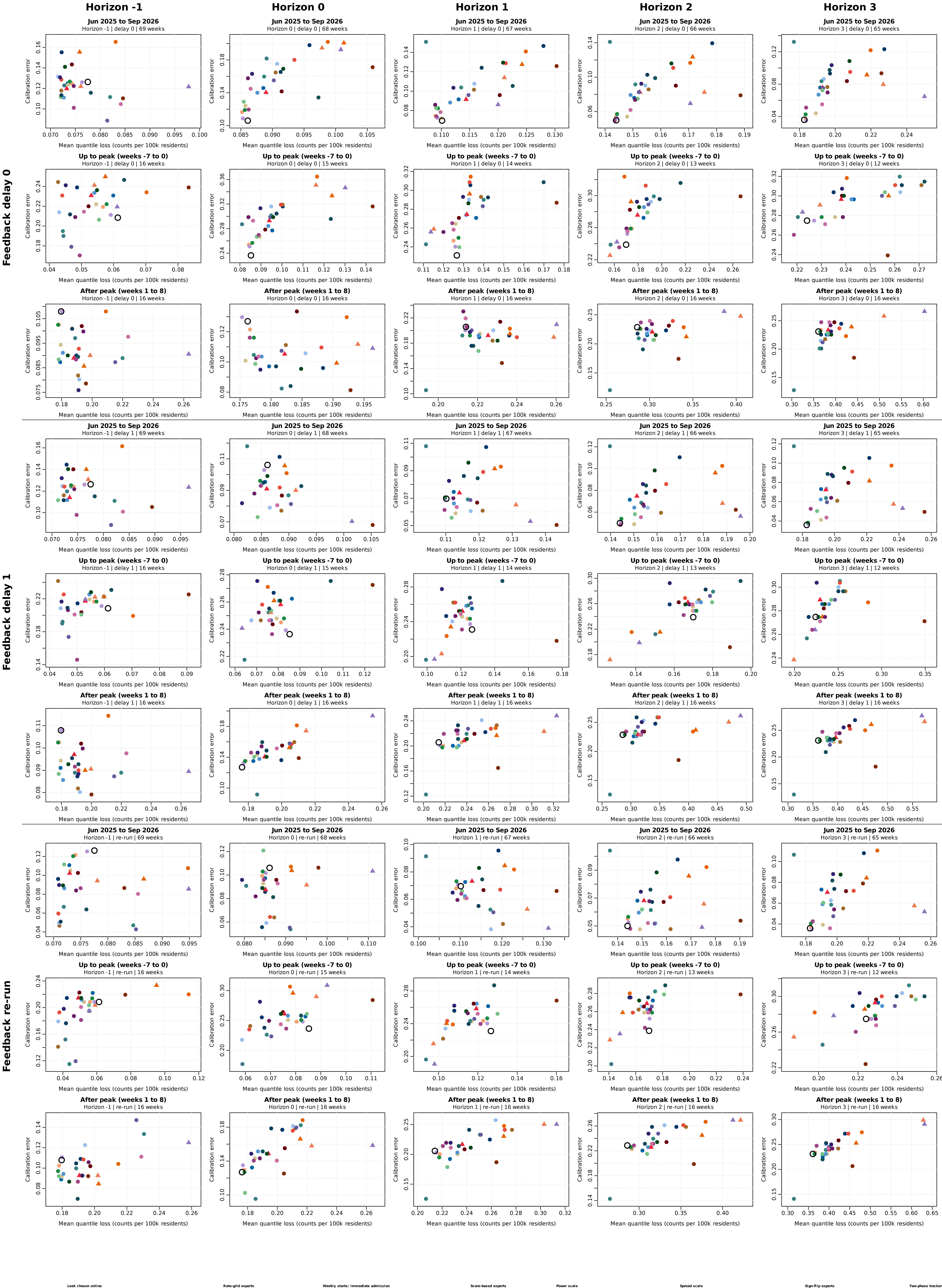
# covid | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.



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# covid | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

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Horizon -1
Horizon 0
Horizon 1
Horizon 2
Horizon 3
Feedback delay 0
Feedback delay 1
Feedback re-run
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks
Horizon 0 | delay 0 | 68 weeks
Horizon 1 | delay 0 | 67 weeks
Horizon 2 | delay 0 | 66 weeks
Horizon 3 | delay 0 | 65 weeks
Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks
Horizon 0 | delay 0 | 16 weeks
Horizon 1 | delay 0 | 16 weeks
Horizon 2 | delay 0 | 16 weeks
Horizon 3 | delay 0 | 16 weeks
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 9 weeks
Horizon 0 | delay 0 | 9 weeks
Horizon 1 | delay 0 | 9 weeks
Horizon 2 | delay 0 | 9 weeks
Horizon 3 | delay 0 | 9 weeks
Horizon -1 | delay 1 | 69 weeks
Horizon 0 | delay 1 | 68 weeks
Horizon 1 | delay 1 | 67 weeks
Horizon 2 | delay 1 | 66 weeks
Horizon 3 | delay 1 | 65 weeks
Horizon -1 | delay 1 | 16 weeks
Horizon 0 | delay 1 | 16 weeks
Horizon 1 | delay 1 | 16 weeks
Horizon 2 | delay 1 | 16 weeks
Horizon 3 | delay 1 | 16 weeks
Horizon -1 | delay 1 | 9 weeks
Horizon 0 | delay 1 | 9 weeks
Horizon 1 | delay 1 | 9 weeks
Horizon 2 | delay 1 | 9 weeks
Horizon 3 | delay 1 | 9 weeks
Horizon -1 | re-run | 69 weeks
Horizon 0 | re-run | 68 weeks
Horizon 1 | re-run | 67 weeks
Horizon 2 | re-run | 66 weeks
Horizon 3 | re-run | 65 weeks
Horizon -1 | re-run | 16 weeks
Horizon 0 | re-run | 16 weeks
Horizon 1 | re-run | 16 weeks
Horizon 2 | re-run | 16 weeks
Horizon 3 | re-run | 16 weeks
Horizon -1 | re-run | 9 weeks
Horizon 0 | re-run | 9 weeks
Horizon 1 | re-run | 9 weeks
Horizon 2 | re-run | 9 weeks
Horizon 3 | re-run | 9 weeks
Calibration error
Mean quantile loss (counts per 100k residents)
Leak chosen online
Leaky MultiQT, leak chosen on the last 20 rows (leaks 0, 0.02, 0.05, 0.1, 0.2, 0.3, 0.5)
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Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
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# covid | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

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Horizon -1
Horizon 0
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Feedback delay 0
Feedback delay 1
Feedback re-run
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks
Horizon 0 | delay 0 | 68 weeks
Horizon 1 | delay 0 | 67 weeks
Horizon 2 | delay 0 | 66 weeks
Horizon 3 | delay 0 | 65 weeks
Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 24 weeks
Horizon 0 | delay 0 | 24 weeks
Horizon 1 | delay 0 | 23 weeks
Horizon 2 | delay 0 | 22 weeks
Horizon 3 | delay 0 | 21 weeks
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 19 weeks
Horizon 0 | delay 0 | 19 weeks
Horizon 1 | delay 0 | 19 weeks
Horizon 2 | delay 0 | 19 weeks
Horizon 3 | delay 0 | 19 weeks
Horizon -1 | delay 1 | 69 weeks
Horizon 0 | delay 1 | 68 weeks
Horizon 1 | delay 1 | 67 weeks
Horizon 2 | delay 1 | 66 weeks
Horizon 3 | delay 1 | 65 weeks
Horizon -1 | delay 1 | 24 weeks
Horizon 0 | delay 1 | 24 weeks
Horizon 1 | delay 1 | 23 weeks
Horizon 2 | delay 1 | 22 weeks
Horizon 3 | delay 1 | 21 weeks
Horizon -1 | delay 1 | 19 weeks
Horizon 0 | delay 1 | 19 weeks
Horizon 1 | delay 1 | 19 weeks
Horizon 2 | delay 1 | 19 weeks
Horizon 3 | delay 1 | 19 weeks
Horizon -1 | re-run | 69 weeks
Horizon 0 | re-run | 68 weeks
Horizon 1 | re-run | 67 weeks
Horizon 2 | re-run | 66 weeks
Horizon 3 | re-run | 65 weeks
Horizon -1 | re-run | 24 weeks
Horizon 0 | re-run | 24 weeks
Horizon 1 | re-run | 23 weeks
Horizon 2 | re-run | 22 weeks
Horizon 3 | re-run | 21 weeks
Horizon -1 | re-run | 19 weeks
Horizon 0 | re-run | 19 weeks
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# covid | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

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Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks
Horizon 0 | delay 0 | 15 weeks
Horizon 1 | delay 0 | 14 weeks
Horizon 2 | delay 0 | 13 weeks
Horizon 3 | delay 0 | 12 weeks
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 15 weeks
Horizon 0 | delay 0 | 15 weeks
Horizon 1 | delay 0 | 15 weeks
Horizon 2 | delay 0 | 15 weeks
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Jun 2025 to Sep 2026
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Horizon -1 | delay 1 | 15 weeks
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Horizon -1 | re-run | 15 weeks
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Leaky MultiQT, leak bank: learned common weights + sharing (H=8)
Rate-grid experts
Rate grid: equal
Rate grid: learned, no sharing (H=8)
Rate grid: learned, no sharing, common weights (H=8)
Weekly starts: immediate admission
Weekly starts: equal (m=0.03)
Weekly starts: most recent 8, learned + sharing (m=0.03, H=8)
Weekly starts: most recent 8, slow + fast, learned + sharing (m=0.03 + 0.3, H=8)
Score-based experts
Score experts: learned weights per level (H=8)
Power scale
Studentized full base ((1 + median)^0.5, adaptive rate)
Power scale: rate grid, learned, common weights (H=8)
Power scale: most recent 8 starts, learned + sharing (m=0.03, H=8)
Spread scale
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Spread scale: rate grid, learned, common weights (H=8)
Spread scale: most recent 8 starts, learned + sharing (m=0.03, H=8)
Sign-flip experts
Sign-flip experts: base, power-scale tracker, mirror (H=8)
Sign-flip experts: base, level-RMS tracker, mirror (H=8)
Sign-flip experts: base, base-width tracker, mirror (H=8)
Two-phase tracker
Two-phase power s
Two-phase power s

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Every tracker learns at the adaptive rate (no tuning); the same settings for every feedback design.

# flu | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

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Horizon 0
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Horizon 3
Feedback delay 0
Feedback delay 1
Feedback re-run
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks
Horizon 0 | delay 0 | 32 weeks
Horizon 1 | delay 0 | 32 weeks
Horizon 2 | delay 0 | 32 weeks
Horizon 3 | delay 0 | 31 weeks
Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks
Horizon 0 | delay 0 | 7 weeks
Horizon 1 | delay 0 | 6 weeks
Horizon 2 | delay 0 | 5 weeks
Horizon 3 | delay 0 | 4 weeks
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Calibration error
Mean quantile loss (counts per 100k residents)
laptive rate)
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Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Every tracker learns at the adaptive rate (no tuning); the same settings for every feedback design.

# flu | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

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Jun 2025 to Sep 2026
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Calibration error
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Two-phase tracker
● Two-phase power s
● Two-phase power s

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Every tracker learns at the adaptive rate (no tuning); the same settings for every feedback design.

# flu | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

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Horizon 1 | delay 1 | 8 weeks
Horizon 2 | delay 1 | 8 weeks
Horizon 3 | delay 1 | 8 weeks
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Horizon 1 | re-run | 32 weeks
Horizon 2 | re-run | 32 weeks
Horizon 3 | re-run | 31 weeks
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Two-phase power s
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Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
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Every tracker learns at the adaptive rate (no tuning); the same settings for every feedback design.

# flu | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

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Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Every tracker learns at the adaptive rate (no tuning); the same settings for every feedback design.

# flu | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

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